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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

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1. Team & Project Overview

Team Name	Hellow World
Selected Domain	Domain 1
Selected SDG Track (1–17)	SDG8 , SDG4
Project / Solution Name	FutureFit(Aligna)
Name (Leader)	LIENG YU HAO
Email & Phone (Leader)	hamyufalanbo@gmail.com
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Many job seekers use job platforms that mainly support searching, filtering and submitting applications. These platforms often show a job title or a match score, but do not clearly explain which evidence supports the result, which requirements are still unknown, or what the person should do next. This creates a practical gap between finding a vacancy and becoming ready for it.

The problem affects students, fresh graduates, career changers, people returning to work, and people whose skills come from informal, part-time, volunteer or project-based experiences. Their relevant abilities may not be represented clearly in a conventional CV. They may also have limited time, budget, location flexibility or access to structured learning.

Employers face a related problem. Job requirements are often written as unstructured text, making it difficult to compare candidates fairly, understand transferable skills and review evidence with appropriate consent.

The resulting problem is avoidable mismatch. Candidates may repeatedly apply without understanding their gaps, while employers may overlook relevant abilities. Opaque recommendations may also cause users to treat a score as a hiring prediction.

Future Fit addresses this gap by connecting self-understanding, job requirements, evidence, learning actions and application reflection in one workflow. Missing information is treated as unconfirmed rather than proof of inability, while final employment decisions remain with people.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Future Fit is a shared seeker–HR workspace that turns career exploration into an explainable improvement loop: understand the user, compare a target role, identify gaps, choose a practical next action, record evidence and review the result.

The platform combines deterministic matching with a bounded generative-AI evidence layer. The shared matching engine compares structured job requirements with confirmed skills, experience and evidence. It calculates requirement coverage and similarity from the same records used by both the seeker and HR views. Each result can therefore show why a requirement is considered met, missing or still unknown. The system does not claim to predict hiring probability, competence or interview success.

An optional server-side Gemini Flash-Lite pipeline processes public workforce and career-related news. It checks the source, publication date, jurisdiction, URL and supporting quotation before publishing a trend item. The AI output is used only as contextual evidence for learning priorities and future-workforce discussion. It does not directly change skill scores or hiring probabilities.

Firebase Authentication and Firestore support identity, private profiles, job records, applications, consented sharing and owner-only learning records. Vercel provides the deployed web application and scheduled news endpoints. The product also provides career exploration, skill evidence, learning routes, work-sample reflection and HR-side candidate comparison.

This approach combines the flexibility of AI summarisation with the transparency and reproducibility of deterministic rules. Users can see what is known, what is missing and what action may help next, instead of receiving an unexplained score.

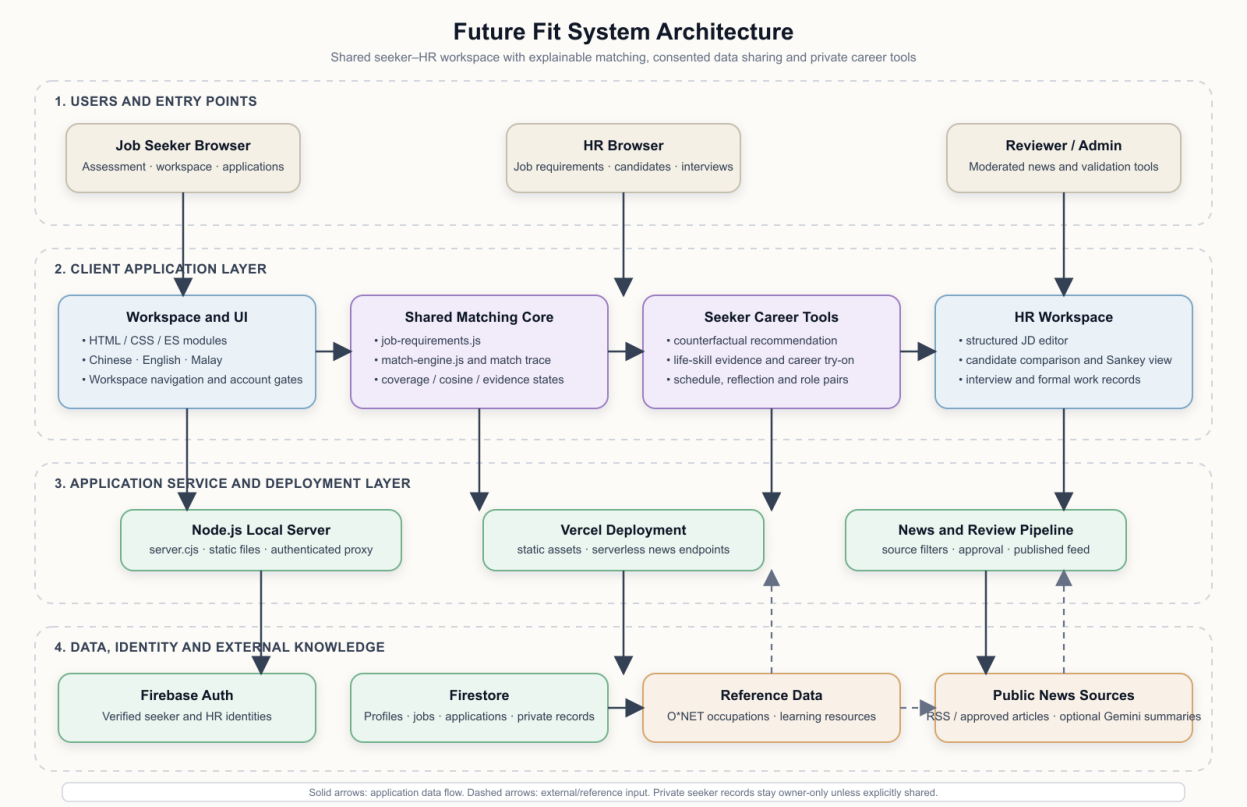
3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Layer	Technology	Purpose
Frontend	HTML, CSS, JavaScript ES Modules	Responsive seeker and HR interfaces
Runtime	Node.js 24.x	Local server and development runtime
Local server	Node.js HTTP server	Static assets and permitted API proxy
Authentication	Firebase Authentication	User identity and role-based access
Database	Cloud Firestore	Profiles, jobs, applications, evidence and private learning records
Deployment	Vercel	Web deployment and scheduled API functions
Matching	Shared deterministic matching engine	Requirement coverage, similarity and match trace
AI layer	Gemini Flash-Lite, optional server-side use	Public news summarisation and trend evidence
External references	O*NET data and public news RSS	Occupation, skills and workforce references
Validation	Node.js test runner and Firestore rules tests	Regression and permission checks



The system uses a shared seeker–HR workspace architecture. Browser-based interfaces provide separate seeker, HR and reviewer entry points while reusing common navigation, structured job requirements and the deterministic matching engine. The Node.js server serves static assets and forwards permitted authenticated requests to the existing production endpoints. Firebase Authentication manages verified identities, while Firestore stores profiles, jobs, applications, consented records and owner-only seeker data. Vercel hosts the deployed application and scheduled news functions. Public occupational references and moderated news evidence remain separate from private user records.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our design-thinking process began with two connected user journeys: a job seeker trying to understand personal strengths and skill gaps, and an HR user trying to describe a role and compare applicants using relevant evidence. We defined the central problem as the lack of an explainable bridge between job discovery, readiness improvement and employer review.

We then prototyped a shared workspace with structured job requirements, skill evidence, transparent matching, career exploration, learning actions, work-sample reflection and consent-controlled application sharing. The seeker and HR views reuse the same job and application records so that explanations remain consistent.

The prototype will be tested through scripted seeker and HR scenarios, including profile completion, job comparison, gap identification, saving a learning action, submitting evidence, reviewing a match explanation and withdrawing consent. We will also test language switching, mobile layouts, private-record boundaries and failure states.

Deployment uses Vercel for the web application and Firebase for authentication and Firestore data. Public news and reference data remain separated from private records. AI-generated summaries require source evidence and review conditions before publication. The current phase demonstrates functional feasibility; professional validity, large-scale user testing and physical-device certification remain future validation work.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 8.5; supporting target SDG 4.4
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	0% of the predefined baseline journeys provide a persistent, evidence-linked next action before using Future Fit. This is an internal prototype baseline, not a national employment statistic.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Percentage of pilot users who can identify a skill gap, understand the reason and save one evidence-linked next action within one session.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	At least 80% of pilot users complete the above flow; at least 95% of same-input seeker and HR match explanations remain consistent.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

No. The submitted solution was developed for this hackathon and does not knowingly include pre-existing team-owned software, product code or commercial assets.

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type	License / Terms
Firebase JavaScript SDK	**Authentication and database SDK**	**Apache-2.0; verify current SDK notice**
Zod	**Runtime validation library**	**MIT**
Mammoth.js	**DOCX document extraction**	**BSD-2-Clause and bundled dependency notices**
PDF.js	**PDF document extraction**	**Apache-2.0**

Name	Type	License / Terms
O*NET data and interest resources	**Occupational and skill reference data**	**O*NET attribution and applicable CC BY 4.0 / Tools Developer License terms**
Google Gemini Flash-Lite	**Optional hosted AI model/API**	**Google AI service terms and applicable usage conditions**
Google News RSS / publisher sources	**Public news inputs**	**Publisher attribution and source terms; not treated as owned content**

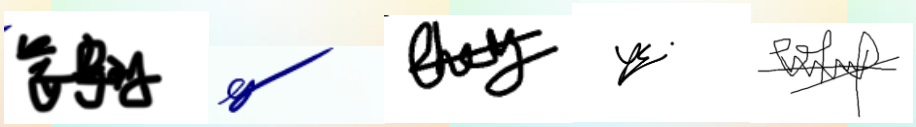
6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No

If yes, please describe the nature of the agreement and which member it concerns:

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature] 

[Date] 24 October 2026